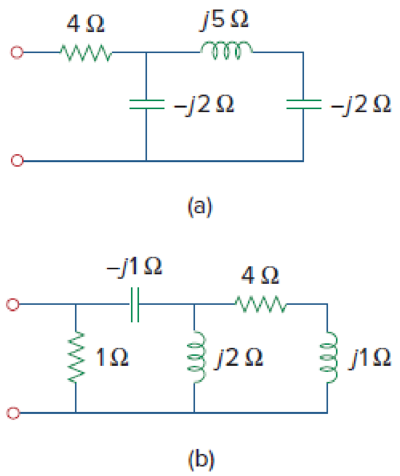
Problem

Obtain the power factor for each of the circuits in Fig. 11.68. Specify each power factor as leading or lagging.

Figure 11.68:



Step-by-step solution

Step 1 of 5

(a)

Refer to circuit in Figure 11.68 (a) in the textbook.

In the circuit, $j5\ \Omega$ inductor and $-j2\ \Omega$ capacitor are connected in series and this series combination is parallel with $-j2\ \Omega$ capacitor. This parallel combination is in series with $4\ \Omega$ resistor.

Calculate the equivalent impedance.

$$\begin{aligned} Z_{eq} &= (j5 - j2) \parallel (-j2) + 4 \\ &= (j3) \parallel (-j2) + 4 \\ &= \frac{(j3)(-j2)}{j3 - j2} + 4 \\ &= 4 - j6 \end{aligned}$$

$$= 7.211\angle -56.31^\circ\ \Omega$$

Step 2 of 5

Calculate the power factor.

$$\begin{aligned} pf &= \cos(-56.31^\circ) \\ &= 0.555 \end{aligned}$$

The power factor angle is negative, so power factor is 0.555 leading.

Therefore, the power factor is $\boxed{0.555\ \text{Leading}}$.

Step 3 of 5

(b)

Refer to circuit in Figure 11.68 (b) in the textbook.

In the circuit, $4\ \Omega$ resistor and $j1\ \Omega$ inductor are connected in series and this series combination is parallel with $j2\ \Omega$ inductor.

Calculate the equivalent impedance.

$$\begin{aligned} Z_{\text{eq1}} &= (4+j1) \| (j2) \\ &= \frac{(4+j1)(j2)}{4+j1+j2} \\ &= \frac{-2+j8}{4+j3} \\ &= 0.64+j1.52\ \Omega \end{aligned}$$

Comments (1)

Anonymous

WHy is the answer positive? Every time I calculate it I get negative numbers.

Step 4 of 5

The simplified circuit is shown in Figure 1.

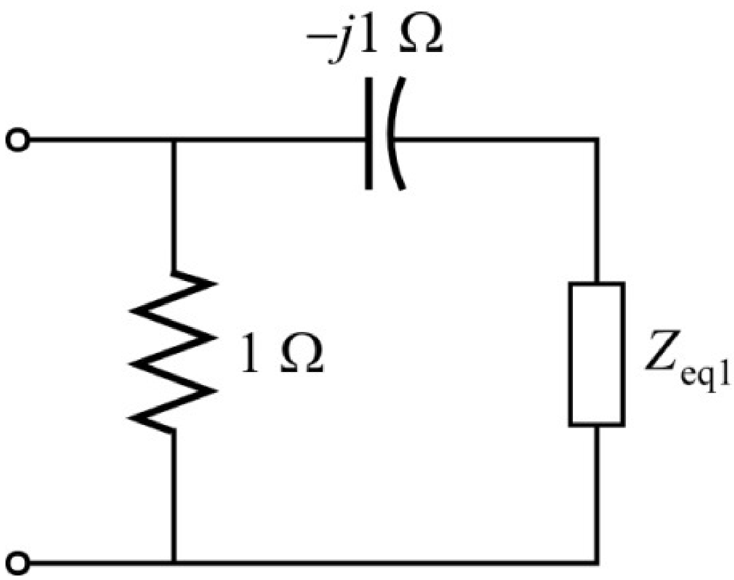


Figure 1

Step 5 of 5

Calculate the equivalent resistance from Figure 1.

$$
\begin{aligned}
Z_{eq} &= (1)\square(-j1+0.64+j1.52) \\
&= (1)\square(0.64+j0.52) \\
&= \frac{(1)(0.64+j0.52)}{1+0.64+j0.52} \\
&= 0.446+j0.1756\ \Omega
\end{aligned}
$$

$$= 0.479\angle 21.5^\circ\ \Omega$$

Calculate the power factor.

$$
\begin{aligned}
pf &= \cos(215^\circ) \\
&= 0.93
\end{aligned}
$$

The power factor angle is positive, so power factor is 0.93 lagging.

Therefore, the power factor is $\boxed{0.93\ \text{Lagging}}$.